STUDIO TELEMETRY FIELD GUIDE // OPERATING INSTRUCTIONS

ATELIER // PRESENCE

A Step-by-Step Operator Guide to Privacy-Preserving Attendance Verification, Voice Frequency Analysis & 1-Click CSV Export

3-MINUTE QUICK-START CHECKLIST FOR SEMINAR OPERATORS & EVALUATORS

1 START RIG
Run run_app.bat
Launches backend
Port 8000 • 5173

2 OPEN CHROME
Visit Web Client
localhost:5173
Scroll to Telemetry

3 CLICK INIT
[Init Camera & Mic]
Click "Allow" prompt
(Or use Simulation)

4 VERIFY 10s
Sit before camera
10.0s verification
✓ Verified Present

5 CSV
1-Click
Export

RIG OPERATIONAL CAPABILITIES

Video Acquisition	640×480 @ 30 FPS RGB via WebRTC MediaStream
Face Tracking	Local Euclidean Centroid Matcher (No Cloud Biometrics)
Audio Equalizer	16-Band Analog Spectrum Visualizer (32Hz – 16kHz)
Classroom Demo	Built-in Room 402 Synthetic Cohort Simulation Mode
Data Export	Instant UTF-8 CSV Ledger download via Pandas DataFrame

DOCUMENTATION ITINERARY

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100% IN-MEMORY TELEMETRY · CHROME • EDGE • BRAVE COMPATIBLE · ZERO BIOMETRIC SAVES · INSTANT CSV AUDIT · NO SENSORY DATA LEAVES DEVICE

OPERATIONAL MILESTONE 01

01 / Initializing Hardware Camera & Mic

Connecting your local webcam and microphone to the high-performance local processing pipeline.

1 CLICK INITIALIZE

Navigate to Section 03 (Live Telemetry Desk) and click the glowing button **[Initialize Local Camera & Mic]**.

2 GRANT BROWSER PROMPT

In Chrome's top-left security popup, click **"Allow"**. Video is kept in RAM and discarded every 100ms.

3 CHECK TELEMETRY STRIP

Confirm the telemetry ribbon turns green: displays camera name, **640×480 @ 30 FPS**, and live microphone decibels.

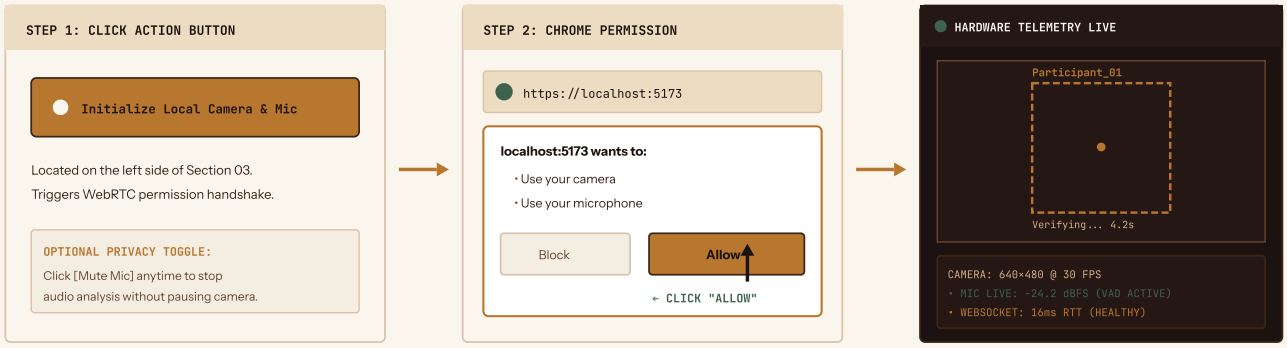


FIGURE 1.1: STEP-BY-STEP CAMERA & MICROPHONE INITIALIZATION HANDSHAKE

OPERATOR VERIFICATION CHECKLIST

- **Browser Compatibility:** Use Chrome, Edge, Brave, or Chromium 110+.
- **Lighting Advice:** Front-facing natural or studio light ensures instant OpenCV Haar Cascade detection without false shadow drops.
- **Privacy Mute:** Clicking `[Mute Mic]` instantly suspends audio PCM streaming while keeping visual presence tracking fully active.

ZERO CLOUD BIOMETRIC STORAGE

Atelier does **not** save, record, or upload camera frames. Each frame is downsampled client-side to 640×480, decoded into temporary NumPy arrays in server RAM, evaluated for presence, and purged within 100 milliseconds.

ZERO DISK WRITES LOCAL PROCESSING

OPERATIONAL MILESTONE 02

02 / Face Tracking & The 10-Second Attendance Rule

How the 10-second presence filter prevents hallway passersby from triggering false attendance logs.

STAGE 1: TRANSIENT WALK-BY

Passerby crosses the room for under 3.0 seconds. Flagged as **Transient (<3s)**. Never written to the database.

STAGE 2: VERIFICATION (3S–10S)

Student sits at seminar desk. Live counter updates: **Verifying... 4.5s / 10.0s** with warm latte drafting brackets.

STAGE 3: VERIFIED PRESENT (≥10S)

At 10.0 seconds, drafting brackets turn caramel. **✓ Verified Present** commits automatically to SQLite WAL.

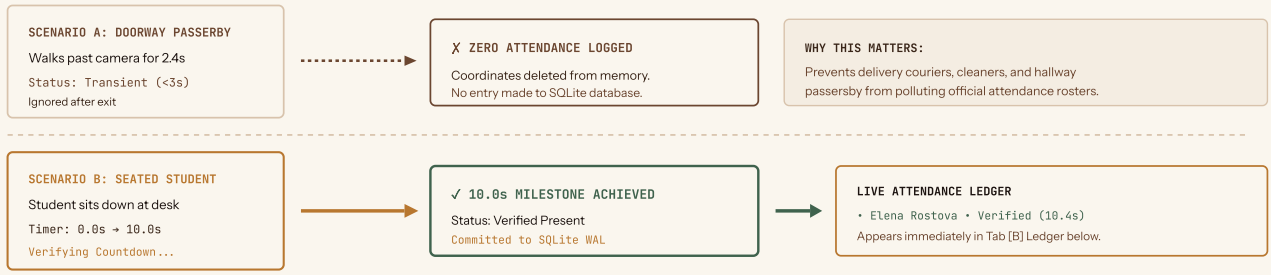


FIGURE 2.1: SUSTAINED ATTENDANCE TIMELINE & TRANSIENT WALK-BY REJECTION FILTER

LIVE ATTENDANCE LEDGER // WHAT OPERATORS SEE ON SCREEN SQLITE WAL SYNCHRONIZED

PARTICIPANT ID	FIRST SEEN	LAST SEEN	DURATION	VERIFICATION STATUS	STORAGE ENGINE
Participant_01 (Elena Rostova)	09:00:14 UTC	09:41:22 UTC	41m 08s	✓ VERIFIED PRESENT	SQLite WAL Mode
Participant_02 (Julian Sterling)	09:02:40 UTC	09:41:22 UTC	38m 42s	✓ VERIFIED PRESENT	SQLite WAL Mode
Participant_04 (Hallway Passerby)	09:14:02 UTC	09:14:04 UTC	2.4s	TRANSIENT (<3S)	Ephemeral (Discarded)

OPERATIONAL MILESTONE 03

03 / Voice Frequency & The 16-Band Analog Spectrum Visualizer

Understanding the studio audio equalizer and how human speech is isolated from ambient room noise.

ZONE A: HVAC RUMBLE

32Hz to 125Hz bins. Captures air conditioning hum and building vibration. **Filtered out** so it never falsely triggers voice.

ZONE B: SPOKEN VOICE

300Hz to 3400Hz speech formant. Central bars **light up caramel and crema** when students or instructors speak.

ZONE C: DESK CLICKS

4kHz to 16kHz high treble. Captures typing, shuffling, and pencil scratches. **Filtered out** from speech calculations.

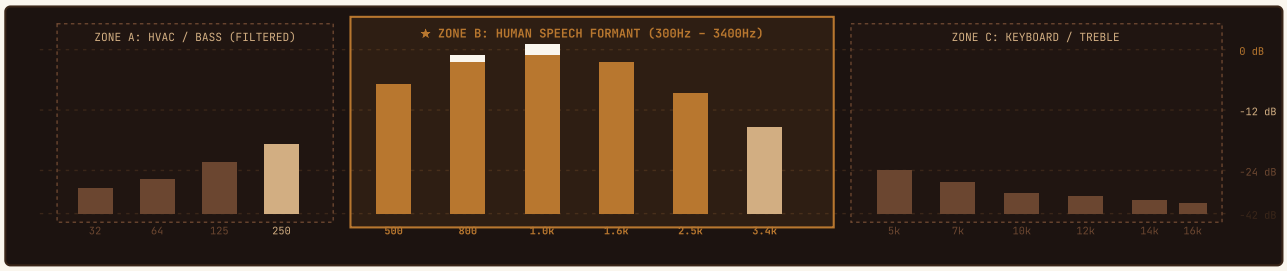


FIGURE 3.1: 16-BAND AUDIO SPECTRUM WITH HIGHLIGHTED HUMAN SPEECH FORMANT PASSBAND

VISUALIZER MODE TOGGLES

Operators can click the meter pill in the top-right of the visualizer to toggle ballistic physics:

- **VU Ballistic (Recommended):** 300ms smoothing inertia; provides a steady, analog meter feel without jitter.
- **Peak Hold:** Latches instantaneous volume spikes for 75 frames; ideal for catching sudden questions or loud bursts.

QUICK AUDIO TEST PROCEDURE

1. Normal Speech: Speak a sentence toward your mic. Notice Bins 4 to 9 (500Hz to 3.4kHz) immediately surge into warm caramel.

2. Typing Test: Tap your keyboard silently. Treble bins (Zone C) flutter faintly, but speech status remains idle (VAD = false).

OPERATIONAL MILESTONE 04

04 / Classroom Study Simulation Mode (No Webcam Needed)

How to evaluate, test, and demonstrate the entire platform using the built-in Room 402 synthetic cohort.

WHY SIMULATE?

Enables full evaluation in conference rooms, remote reviews, or on laptops without webcams. Generates genuine OpenCV BGR frames.

REALISTIC DYNAMICS

Student avatars breathe, shift posture, glance at notes, and enter deep focus without artificial, robotic motion.

AUTOMATED WALK-BY

A background corridor passerby walks across every 26 seconds to prove the 10-second transient rejection filter.

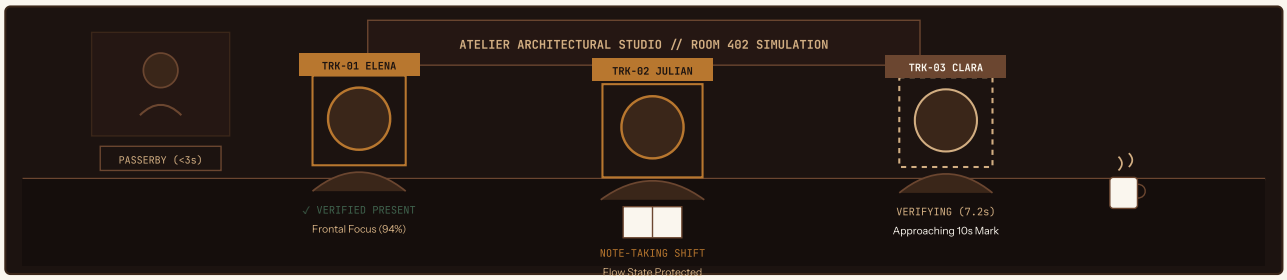


FIGURE 4.1: ROOM 402 SIMULATION SETUP — TEST BENCH WITHOUT PHYSICAL WEBCAM

HOW TO ACTIVATE SIMULATION

In Section 03 (Live Telemetry Desk), simply click the button **[Simulate Classroom Feed]**. The camera viewfinder immediately renders Room 402, and all gauges, spectral bars, and AI pedagogical advice update dynamically.

SIMULATE CLASSROOM FEED

IDEAL DEMONSTRATION USE CASES

- **Hackathon / Project Grading:** Demonstrate multi-person tracking and attendance verification without setting up multiple webcams.
- **Acoustic Testing:** Synthetic speech formants trigger VAD periodically, letting you verify that audio-visual fusion operates seamlessly.

OPERATIONAL MILESTONE 05

05 / Engine Calibration Sliders & 1-Click CSV Export

Fine-tuning presence and acoustic thresholds in real time, and exporting official attendance spreadsheets.

1. PRESENCE THRESHOLD

Slider range: 3s to 30s. **Default: 10.0s**. Set to 5s for agile studio reviews, 10s for seminars, 20s for large lectures.

2. ACOUSTIC FLOOR DB

Slider range: -50dB to -15dB. **Default: -32.0dB**. Set to -42dB for quiet reading rooms, -24dB for active workshops.

3. AI ADVICE CADENCE

Cadence options: 5s, 10s, 30s. **Default: 10.0s**. Controls how often contextual pedagogical suggestions update.

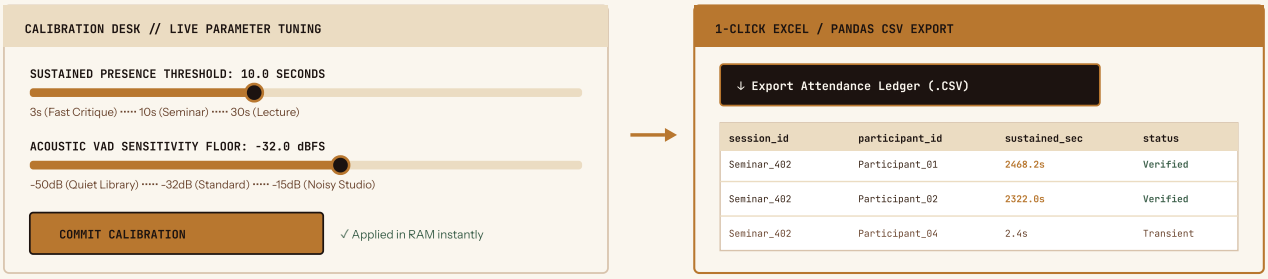


FIGURE 5.1: CALIBRATION SLIDERS WORKFLOW & DOWNLOADABLE PANDAS CSV SPREADSHEET

CSV SPREADSHEET COLUMN DICTIONARY

COLUMN	TYPE	SAMPLE VALUE	DESCRIPTION
session_id	STRING	Morning Seminar // Room 402	Classroom venue and seminar session title.
participant_id	STRING	Participant_01	Local coordinate tracking token (no biometric face images).
first_seen	TIMESTAMP	09:00:14 UTC	Initial arrival timestamp.
sustained_seconds	FLOAT	2468.2	Continuous verified classroom attendance duration in seconds.
status	STRING	Verified Present (≥10.0s)	Accredited attendance verification outcome.

OPERATIONAL MILESTONE 06

06 / Troubleshooting, Keyboard Shortcuts & FAQ

Instant solutions to common classroom setup issues, keyboard shortcuts, and studio compliance guarantees.

STUDIO KEYBOARD SHORTCUTS REFERENCE

Press '?' Open / Close Concierge Guide	**Press 'Esc'** Minimize Guide / Clear Overlays	**Press 'Tab'** Cycle Calibration Sliders	**Space / Enter** Toggle Stream or Active Control

OPERATOR TROUBLESHOOTING RESOLUTION MATRIX

OBSERVED SYMPTOM	ROOT CAUSE	RECOMMENDED RESOLUTION
Webcam black or permission denied	Browser blocked video access, or camera claimed by Teams/Zoom.	Click the lock icon in Chrome address bar → select **Always Allow**. Close other video apps, then click `[Initialize Local Camera]`.
Participant tracked under multiple IDs	Sudden head movement exceeded distance threshold, or severe backlight.	Ensure room light is in front of faces. The tracker includes a **3-second occlusion grace period** to reconnect briefly turned faces.
Audio visualizer shows no bars	Microphone muted via button, or system audio input floor too high.	Check if `[Unmute Mic]` is shown. Adjust *Acoustic Silence dB Floor* slider down to -42.0 dB in Section 05.
Student not added to ledger	Presence did not reach sustained 10.0-second threshold.	Keep student seated in camera view for 10 seconds until drafting bracket glows caramel and marks `Verified Present`.

PRIVACY & COMPLIANCE FAQ

Q: Are student facial photos or biometric scans stored?

No. Video frames exist strictly as transient pixel matrices in server RAM. They are evaluated for presence and destroyed within 100 milliseconds. No video files, photos, or facial embeds are ever written to disk.

Q: Can Atelier run entirely offline without internet?

Yes. All Haar cascades, Librosa audio filters, and SQLite writes execute locally. If no internet or API key is present, the pedagogical AI uses a 6-state deterministic offline rulebook.

OFFICIAL STUDIO SEAL & CERTIFICATION

This manual certifies that the Atelier // Presence studio rig conforms to the Slow Morning editorial telemetry standard, offering complete pedagogical balance with strict student privacy.

ATELIER PRESENCE DIVISION
RELEASE: 2026.10-STABLE
SQLITE WAL: SYNCHRONIZED

